Algebra Monsoon 2026 Lecture 19 Solvable groups

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Solvable groups

Let G be a group.

For subgroups $A, B \subset G$, let (A, B) be the subgroup of G generated by all commutators

$$xyx^{-1}y^{-1} \quad (x \in A, y \in B).$$

Recall that $D(G) = (G, G)$ is a characteristic subgroup of G.

Define $\begin{cases} D^0(G) = G \\ D^{n+1}(G) = (D^n(G), D^n(G)) \end{cases}$ for $n \geq 0$.

This is a decreasing sequence of subgroups of G

this is the derived series of G:

$$\cdots \subset D^2(G) \subset D^1(G) \subset D^0(G) = G$$

Definition We say that G is solvable if $D^n(G) = \{e\}$ for some $n \geq 1$.

Proposition The following are equivalent:

1) G is solvable,

2) There is a decreasing sequence

$$\{e\} = G_m \subset \cdots \subset G_1 \subset G_0 = G$$

of normal subgroups of G such that G_i/G_{i+1} is commutative for every $i \in [0, m]$

3) There is a decreasing filtration

$$\{e\} = G_n \subset \cdots \subset G_1 \subset G_0 = G$$

such that G_i/G_{i+1} is commutative for all $i \in [0, n]$

(so we don't require the G_i to be normal in G)

1) $\Rightarrow$ 2) Take $G_i = D^i(G)$ for every $i \geq 0$.
As $D(G)$ is a characteristic subgroup of G, $D^i(G)$ is normal in G for every $i \geq 0$.

2) $\Rightarrow$ 3) Trivial.

3) $\Rightarrow$ 1) By induction on k, we see that $D^k(G) \subset G_k$ for every $k \geq 0$,
Therefore $D^n(G) = \{e\}$ □

Proposition If a group G is solvable and simple, then it is cyclic of prime order.

Proof. Consider $D(G)$. It is a normal subgroup of G. If we have $D(G) = G$, we will have $D^n(G) = G$ for every $n \geq 1$ and G will not be solvable. Therefore $D(G) = \{1\}$

As a result, G is commutative.

Pick any element $a \neq e$ in G.

The subgroup $\langle a \rangle$ is normal in G (because G is commutative), so $\langle a \rangle = G$ (because G is simple).

Therefore G is cyclic. Moreover G has prime order because cyclic groups of composite order are not simple.

□

Proposition Let $1 \longrightarrow N \hookrightarrow G \xrightarrow{\pi} Q \longrightarrow 1$ be an exact sequence of groups. We have

G is solvable $\iff$ N and Q are solvable.

Proof. $\longrightarrow$: If $\{e\} = G_n \subset \cdots \subset G_1 \subset G_0 = G$ is a subnormal filtration on G such that

is a filtration $\{e\} = G_r \subset \cdots \subset G_1 \subset G_0 = G$ such that
G_i/G_{i+1} is commutative for every $i \in [1, n-1]$,
then $N_i = N \cap G_i$ give such a filtration on N and $Q_i = \pi_*(G_i)$ give such a filtration on Q.

Conversely, given filtrations

$$\{e\} = N_r \subset \cdots \subset N_1 \subset N_0 = N$$

and $\{e\} = Q_s \subset \cdots \subset Q_1 \subset Q_0 = Q$,

take the filtration

$$\{e\} \subset N_r \subset \cdots \subset N_1 \subset \pi^*(Q_s) \subset \cdots \subset \pi^*(Q_1) \subset G$$

on G.

We could have defined solvable groups by declaring that

1) Commutative groups are solvable,
2) extensions of solvable groups are solvable.

Here, a group G is said to be an extension of a group Q by a

group N if there is an exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow 1.$$

- $\mathbb{Z}/6\mathbb{Z}$ is an extension of $\mathbb{Z}/2\mathbb{Z}$ by $\mathbb{Z}/3\mathbb{Z}$

- S_3 ⟶ _ _ _ .